

Computing Theory and Practices (2026)

Course code 24PY902
M2U1

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August 1, 2026

M2U1 Plan

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After completing this lecture, you will be able to

- Explain the concept of an artificial atom.
- Compare natural atoms and superconducting artificial atoms.
- Describe how Josephson junction circuits create discrete energy levels.
- Explain how artificial atoms function as qubits.
- Identify the advantages of engineered quantum circuits.

What is an Artificial Atom?

A natural atom has discrete electronic energy levels because electrons are bound to a nucleus.

A superconducting circuit containing one or more Josephson junctions can also possess discrete energy levels.

Definition

An **artificial atom** is a man-made superconducting circuit engineered to exhibit quantized energy levels similar to those of a natural atom.

Natural Atoms vs Artificial Atoms

Natural Atom	Artificial Atom
Occurs naturally	Fabricated using microelectronics
Electrons bound to nucleus	Cooper pairs in superconducting circuit
Fixed energy levels	Engineered energy levels
Atomic dimensions	Micrometre-scale device
Properties fixed by nature	Properties can be designed and tuned

Role of the Josephson Junction

The Josephson junction provides the nonlinear behavior required for quantum operation.

A superconducting circuit containing

- Josephson junctions,
- capacitors,
- inductors,

forms a quantum circuit with discrete energy levels.

The two lowest energy levels are used as

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

Artificial Atom as a Qubit

The superconducting circuit behaves like a two-level quantum system.

Quantum information is stored in

$$|0\rangle$$

and

$$|1\rangle.$$

Microwave pulses are used to

- initialize,
- manipulate,
- read out

the qubit state.

Advantages of Artificial Atoms

Artificial atoms offer several engineering advantages.

- Energy levels can be designed.
- Device parameters are tunable.
- Compatible with semiconductor fabrication.
- Easily integrated into quantum circuits.
- Scalable to multi-qubit processors.

These advantages make superconducting circuits attractive for large-scale quantum computing.

Artificial atoms are used in

- Superconducting quantum processors
- Quantum simulation
- Circuit Quantum Electrodynamics (cQED)
- Quantum communication
- Quantum sensing
- Quantum error-correction research

They form the basis of many gate-based quantum computing platforms.

Worked Example

Why is a Josephson junction called an “artificial atom”?

Answer

- It possesses discrete quantum energy levels.
- These energy levels can represent qubit states.
- Microwave radiation induces transitions between the levels, similar to optical transitions in natural atoms.
- The energy spectrum is engineered rather than occurring naturally.

Summary and Concept Check

Summary

- Artificial atoms are engineered superconducting circuits.
- Josephson junctions create quantized energy levels.
- The two lowest energy levels encode qubits.
- Microwave pulses manipulate quantum states.
- Artificial atoms are the building blocks of superconducting quantum processors.

Concept Check

- 1 What is an artificial atom?
- 2 How does an artificial atom differ from a natural atom?
- 3 Why is a Josephson junction essential for superconducting qubits?
- 4 Why are artificial atoms well suited for scalable quantum computing?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of a semiconductor quantum dot.
- Describe quantum confinement and discrete energy levels.
- Explain how quantum dots function as qubits.
- Identify common materials used for quantum dots.
- Discuss the advantages and challenges of quantum-dot quantum computers.

What is a Quantum Dot?

A quantum dot is a nanoscale semiconductor structure that confines electrons (or holes) in all three spatial dimensions.

Because of this confinement, electrons can occupy only discrete energy levels.

Definition

A quantum dot is often called an **artificial atom** because its energy spectrum resembles that of a natural atom.

Quantum Confinement

When the size of a semiconductor crystal becomes comparable to the electron's de Broglie wavelength, continuous energy bands split into discrete energy levels.

Key consequences

- Quantized energy spectrum
- Size-dependent electronic properties
- Tunable optical properties

Smaller quantum dots generally exhibit larger energy spacing.

Quantum Dot as a Qubit

A single electron can be confined inside a quantum dot.

Qubit information may be encoded using

- Electron spin
- Charge state
- Valley state (in some materials)

The two logical states are represented as

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

Materials and Fabrication

Common semiconductor materials include

- Silicon (Si)
- Germanium (Ge)
- Gallium Arsenide (GaAs)
- Indium Arsenide (InAs)

Quantum dots are fabricated using

- Lithographic patterning
- Electrostatic gate confinement
- Self-assembled nanostructures

Advantages and Challenges

Advantages

Compatible with CMOS technology

Small device size

Potentially scalable

Long spin coherence (Si)

Challenges

Fabrication precision required

Sensitive to charge noise

Low-temperature operation

Complex qubit control

Quantum dots are being explored for

- Spin-based quantum computing
- Quantum communication
- Single-photon sources
- Quantum sensing
- Nanoelectronics
- Quantum simulation

Silicon quantum dots are attractive because they leverage existing semiconductor manufacturing technology.

Worked Example

Why is a quantum dot often referred to as an “artificial atom”?

Answer

- Electrons are confined to a very small region.
- The confinement produces discrete energy levels.
- These energy levels resemble those found in natural atoms.
- The structure is engineered rather than naturally occurring.

Summary and Concept Check

Summary

- Quantum dots confine charge carriers in three dimensions.
- Quantum confinement produces discrete energy levels.
- Electron spin or charge can encode qubits.
- Silicon quantum dots are compatible with modern semiconductor fabrication.
- Quantum dots are a promising platform for scalable quantum computing.

Concept Check

- 1 What is meant by quantum confinement?
- 2 Why are quantum dots called artificial atoms?
- 3 Which physical property is commonly used to encode a qubit in a quantum dot?
- 4 Why are silicon quantum dots attractive for large-scale quantum computers?

After completing this lecture, you will be able to

- Explain the concept of intrinsic spin.
- Describe the quantum states of a spin-1/2 particle.
- Represent spin states using Dirac notation.
- Relate spin states to qubits.
- Explain the significance of spin-1/2 systems in quantum computing.

What is Spin?

Spin is an intrinsic quantum property of elementary particles.

Unlike classical rotation, quantum spin does not correspond to a particle physically spinning about its axis.

Examples of spin-1/2 particles include

- Electron
- Proton
- Neutron

Spin is a form of angular momentum that exists even when the particle is at rest.

Spin-1/2 Quantum States

For a spin-1/2 particle, measurement along the z-axis yields only two possible outcomes.

These basis states are

$$|+\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |-\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

They are often denoted as

$$|\uparrow\rangle \quad \text{and} \quad |\downarrow\rangle.$$

Spin States as Qubits

A spin-1/2 particle naturally forms a two-level quantum system.

The logical qubit states are identified as

$$|0\rangle \equiv |\uparrow\rangle, \quad |1\rangle \equiv |\downarrow\rangle.$$

A general spin state is

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Measurement of Spin

Spin is measured using a magnetic field, as demonstrated in the Stern–Gerlach experiment.

Possible measurement outcomes are

$$+\frac{\hbar}{2} \quad \text{or} \quad -\frac{\hbar}{2}.$$

The probabilities of these outcomes are determined by the amplitudes of the quantum state.

Pauli Spin Matrices

The spin operators are represented by the Pauli matrices.

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

These operators describe spin measurements along the x-, y-, and z-axes.

Applications in Quantum Computing

Spin-1/2 systems are used in

- Electron spin qubits
- Semiconductor quantum dots
- Nitrogen-vacancy (NV) centers
- Nuclear magnetic resonance (NMR)
- Spin-based quantum sensors

Their long coherence times make them promising candidates for quantum information processing.

Worked Example

An electron is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle).$$

Question:

What are the probabilities of measuring

$$|0\rangle \quad \text{and} \quad |1\rangle?$$

Answer:

$$P(0) = \frac{1}{2}, \quad P(1) = \frac{1}{2}.$$

Summary and Concept Check

Summary

- Spin is an intrinsic quantum property.
- Spin-1/2 particles have two allowed measurement outcomes.
- Spin states form a natural qubit.
- Pauli matrices describe spin measurements.
- Spin qubits are widely used in quantum technologies.

Concept Check

- 1 What is meant by intrinsic spin?
- 2 Why is a spin-1/2 particle a natural two-level system?
- 3 What are the possible outcomes of a spin measurement along the z -axis?
- 4 Which Pauli matrix corresponds to a spin measurement along the x -axis?

Learning Outcomes

After completing this lecture, you will be able to

- Explain photon polarization as a qubit.
- Identify common polarization basis states.
- Represent polarization states using Dirac notation.
- Describe optical components used to manipulate polarization.
- Explain the role of polarization qubits in quantum communication.

Photon Polarization

A photon is the quantum of electromagnetic radiation.

Its polarization describes the orientation of the electric field.

Common polarization states include

- Horizontal (H)
- Vertical (V)
- Diagonal (D)
- Anti-diagonal (A)
- Right Circular (R)
- Left Circular (L)

Polarization as a Qubit

The computational basis is chosen as

$$|H\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |V\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A general polarization qubit is

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Superposition of Polarization States

Diagonal polarization is

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}.$$

Anti-diagonal polarization is

$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}.$$

These are examples of quantum superposition.

Manipulating Polarization Qubits

Common optical components include

- Polarizers
- Half-wave plates
- Quarter-wave plates
- Polarizing beam splitters
- Single-photon detectors

These devices prepare, transform, and measure polarization states.

Advantages and Challenges

Advantages

- Long-distance transmission
- Low environmental interaction
- Ideal for quantum communication
- Room-temperature operation

Challenges

- Photon loss
- Single-photon generation
- Efficient detection
- Scalable photon-photon interactions

Applications

Photon polarization qubits are widely used in

- Quantum key distribution (QKD)
- Quantum teleportation
- Quantum networking
- Linear optical quantum computing
- Satellite quantum communication

Their low interaction with the environment makes them ideal carriers of quantum information.

Worked Example

A photon is prepared in the state

$$|\psi\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}.$$

Question:

What is the probability of measuring

$$|H\rangle \quad \text{and} \quad |V\rangle?$$

Answer

$$P(H) = \frac{1}{2}, \quad P(V) = \frac{1}{2}.$$

Summary and Concept Check

Summary

- Photon polarization provides a natural two-level quantum system.
- Horizontal and vertical polarizations form the computational basis.
- Superposition enables arbitrary polarization states.
- Optical components manipulate polarization qubits.
- Polarization qubits are fundamental to quantum communication.

Concept Check

- 1 Why is photon polarization suitable for representing a qubit?
- 2 What are the computational basis states for polarization qubits?
- 3 Which optical component converts linear to circular polarization?
- 4 Why are photons preferred for quantum communication?

Learning Outcomes

After completing this lecture, you will be able to

- Explain why arbitrary quantum gates are difficult to implement.
- Describe the concept of a universal quantum gate set.
- State the Solovay–Kitaev Theorem qualitatively.
- Explain how quantum compilers approximate quantum circuits.
- Recognize the importance of the theorem in quantum computing.

Motivation

Quantum algorithms often require arbitrary unitary operations.

However, real quantum hardware provides only a finite set of physical gates.

Question:

Can a finite set of quantum gates perform any quantum computation?

The Solovay–Kitaev Theorem answers this question.

Universal Quantum Gate Sets

A gate set is called **universal** if any unitary operation can be approximated to arbitrary accuracy.

Examples include

- Hadamard (H)
- Phase (S)
- T Gate
- CNOT Gate

Complex quantum circuits can be built from these elementary gates.

Statement of the Solovay–Kitaev Theorem

Qualitative Statement

Any quantum gate can be efficiently approximated using a finite universal gate set.

The approximation error can be made arbitrarily small by using a sufficiently long sequence of gates.

Thus, a finite gate library is sufficient for universal quantum computation.

Why is the Theorem Important?

Without the theorem,
every quantum computer would require infinitely many different physical gates.

With the theorem,

- only a finite gate library is needed,
- hardware becomes practical,
- quantum programs become portable across different devices.

Role in Quantum Compilation

Quantum programmers write high-level algorithms.

Quantum compilers translate these algorithms into sequences of hardware-supported gates.

Example

$$U \implies H \rightarrow T \rightarrow H \rightarrow S \rightarrow \dots$$

The compiler uses gate approximations based on the Solovay–Kitaev Theorem.

Applications

The theorem is fundamental to

- Quantum compiler design
- Fault-tolerant quantum computing
- Gate synthesis
- Quantum circuit optimization
- Universal quantum computation

Modern quantum software frameworks rely on gate decomposition techniques inspired by this theorem.

Worked Example

Suppose a quantum processor supports only

H , T , and CNOT.

Question:

Can it implement an arbitrary single-qubit rotation exactly?

Answer:

Not always exactly.

However,

using sufficiently long sequences of these gates,
the desired operation can be approximated to any required accuracy.

Summary and Concept Check

Summary

- Real quantum hardware provides only a finite gate library.
- Universal gate sets can approximate arbitrary unitary operations.
- The Solovay–Kitaev Theorem guarantees efficient approximation.
- Quantum compilers use this principle to generate executable circuits.
- The theorem is a cornerstone of scalable quantum computing.

Concept Check

- 1 Why is a universal gate set important?
- 2 What problem does the Solovay–Kitaev Theorem solve?
- 3 Why are arbitrary quantum gates approximated rather than implemented directly?
- 4 Name two areas of quantum computing where this theorem is applied.

After completing this lecture, you will be able to

- Define quantum advantage.
- Distinguish between quantum advantage and quantum supremacy.
- Identify problems where quantum computers may outperform classical computers.
- Recognize current limitations of quantum hardware.
- Explain the future impact of quantum technologies.

What is Quantum Advantage?

Quantum advantage refers to the ability of a quantum computer to solve a practical problem more efficiently than the best known classical approach.

It does **not** imply that quantum computers are faster for every problem.

Instead, they offer advantages for selected classes of computational tasks.

Quantum Advantage vs Quantum Supremacy

Quantum Advantage

Focuses on useful practical problems

Emphasizes real-world applications

Relevant to industry and science

Quantum Supremacy

Focuses on demonstrating computational capability

Often uses specially designed benchmark problems

Primarily a research milestone

Quantum advantage is the broader and more application-oriented concept.

Where Can Quantum Computers Help?

Potential application areas include

- Quantum chemistry
- Drug discovery
- Materials science
- Optimization
- Machine learning
- Cryptography
- Financial modelling

These problems often involve computational complexity that challenges classical computers.

Why Do Quantum Computers Offer an Advantage?

Quantum computers exploit

- Superposition
- Entanglement
- Interference
- Quantum parallelism

These uniquely quantum resources enable algorithms that can outperform classical methods for specific tasks.

Current Challenges

Present-day quantum computers face several limitations.

- Decoherence
- Noise
- Limited qubit counts
- Gate errors
- Error correction overhead

Most existing devices belong to the **Noisy Intermediate-Scale Quantum (NISQ)** era.

Examples of Quantum Advantage

Examples of quantum algorithms include

- Shor's algorithm (integer factorization)
- Grover's algorithm (database search)
- Variational Quantum Eigensolver (VQE)
- Quantum Approximate Optimization Algorithm (QAOA)

Many of these algorithms continue to be explored on modern quantum hardware.

Worked Example

A logistics company must optimize delivery routes for thousands of vehicles.

Question:

Why might a quantum computer be useful?

Answer:

Optimization problems can become computationally expensive for classical computers.

Quantum optimization algorithms may explore solution spaces more efficiently, although practical performance depends on the algorithm and available hardware.

Summary and Concept Check

Summary

- Quantum advantage means solving selected practical problems more efficiently than the best known classical methods.
- It differs from quantum supremacy, which demonstrates computational capability on benchmark tasks.
- Superposition, entanglement, and interference are key quantum resources.
- Current quantum hardware remains limited by noise and decoherence.
- Continued advances in hardware, algorithms, and error correction are expected to expand the range of practical applications.

Concept Check

- 1 What is meant by quantum advantage?
- 2 How does quantum advantage differ from quantum supremacy?
- 3 Name two application areas where quantum computers may provide an advantage.
- 4 What are two major challenges facing current quantum computers?

Learning Outcomes

After completing this lecture, you will be able to

- Define classical and quantum correlations.
- Distinguish between classical and quantum correlations.
- Explain how quantum correlations arise.
- Describe the relationship between quantum correlations and entanglement.
- Identify applications of quantum correlations.

What are Correlations?

Correlation measures how the outcomes of two or more systems are related.

Examples of classical correlations include

- Matching socks in a pair
- Outcomes of linked mechanical systems
- Weather measurements at nearby locations

Classical correlations arise from shared preparation or common causes.

Classical vs Quantum Correlations

Classical Correlations

Explained by classical physics
Based on predetermined properties
No superposition required
Limited by classical probability

Quantum Correlations

Require quantum mechanics
Depend on quantum states
Can involve superposition
May exceed classical limits

Origin of Quantum Correlations

Quantum correlations arise because quantum systems can exist in superposition and interact coherently.

Important ingredients include

- Superposition
- Interference
- Shared quantum states
- Entanglement (in many cases)

Not every quantum correlation requires entanglement, but all entangled systems exhibit quantum correlations.

Example: Bell State

Consider the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$

Measurements on the two qubits are strongly correlated.

These correlations cannot be fully explained using classical models.

Quantum Correlations and Entanglement

Entanglement is one important source of quantum correlations.

- Entangled states always exhibit quantum correlations.
- Some mixed quantum states can display quantum correlations without being maximally entangled.
- Quantum discord is another measure of non-classical correlations.

Entanglement is therefore a subset of the broader concept of quantum correlations.

Applications

Quantum correlations enable

- Quantum communication
- Quantum teleportation
- Superdense coding
- Quantum cryptography
- Quantum sensing
- Quantum computing

These applications exploit non-classical relationships between quantum systems.

Worked Example

Two qubits are prepared in the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Question:

If the first qubit is measured to be $|0\rangle$, what is the expected outcome for the second qubit?

Answer:

The second qubit is also found in the state

$$|0\rangle.$$

The outcomes are perfectly correlated in the computational basis.

Summary and Concept Check

Summary

- Correlations describe relationships between measurement outcomes.
- Quantum correlations extend beyond classical correlations.
- Superposition and coherent quantum evolution give rise to non-classical correlations.
- Entanglement is a powerful source of quantum correlations.
- Quantum correlations underpin many quantum information technologies.

Concept Check

- 1 What distinguishes quantum correlations from classical correlations?
- 2 Name two physical mechanisms that give rise to quantum correlations.
- 3 Does every quantum correlation require entanglement?
- 4 Give two applications that rely on quantum correlations.

Learning Outcomes

After completing this lecture, you will be able to

- Define quantum entanglement.
- Distinguish separable and entangled quantum states.
- Identify Bell states.
- Explain the physical significance of entanglement.
- Describe applications of entanglement in quantum technologies.

What is Quantum Entanglement?

Quantum entanglement is a phenomenon in which two or more quantum systems share a joint state that cannot be expressed as independent states of the individual systems.

When particles are entangled, the outcome of one measurement is strongly correlated with the outcome of another, even when the particles are spatially separated.

Entanglement is a uniquely quantum phenomenon with no classical analogue.

Separable and Entangled States

A separable (product) state can be written as

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle.$$

An entangled state cannot be factorized into the product of individual qubit states.

Example

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This Bell state is entangled.

Bell States

The four Bell states form an orthonormal basis for two-qubit systems.

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}$$

$$|\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

Bell states represent maximally entangled two-qubit states.

Properties of Entanglement

Important characteristics include

- Strong non-classical correlations.
- Measurement outcomes are correlated.
- Entanglement cannot be created by local operations alone.
- Entanglement does not enable faster-than-light communication.
- It is a valuable resource for quantum information processing.

Creating Entanglement

A simple quantum circuit can generate a Bell state.

Steps

- 1 Prepare the state

$$|00\rangle.$$

- 2 Apply a Hadamard gate to the first qubit.
- 3 Apply a CNOT gate.

The resulting state is

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Applications of Entanglement

Entanglement is central to

- Quantum teleportation
- Superdense coding
- Quantum key distribution
- Quantum error correction
- Quantum networking
- Quantum computing algorithms

Many quantum advantages arise from the controlled creation and manipulation of entangled states.

Worked Example

Consider the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Question:

If the first qubit is measured to be

$$|1\rangle,$$

what is the state of the second qubit?

Answer:

The second qubit is also measured in the state

$$|1\rangle.$$

The measurement outcomes are perfectly correlated in the computational basis.

Summary and Concept Check

Summary

- Entanglement is a uniquely quantum correlation between particles.
- Entangled states cannot be written as products of individual states.
- Bell states are maximally entangled two-qubit states.
- Entanglement is an essential resource for quantum information processing.
- It enables protocols such as teleportation and superdense coding.

Concept Check

- 1 What distinguishes an entangled state from a separable state?
- 2 Name the four Bell states.
- 3 Can entanglement be used for faster-than-light communication? Explain briefly.
- 4 Give two applications of quantum entanglement.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the purpose of a Turing machine.
- Identify the main components of a Turing machine.
- Describe the operation of a Turing machine.
- Understand the concept of computability.
- Relate classical Turing machines to quantum computation.

What is a Turing Machine?

A Turing machine is an abstract mathematical model of computation proposed by Alan Turing in 1936.

It provides a simple yet powerful model for describing how algorithms are executed.

A Turing machine forms the theoretical foundation of modern computer science.

Components of a Turing Machine

A classical Turing machine consists of

- An infinite tape divided into cells
- A read/write head
- A finite set of states
- An input alphabet
- A tape alphabet
- A transition function

The tape serves as the machine's memory.

How Does a Turing Machine Work?

At each computation step, the machine

- 1 Reads the current tape symbol.
- 2 Changes its internal state.
- 3 Writes a new symbol (if required).
- 4 Moves the tape head left or right.

This process continues until a halting state is reached.

State Transition Function

The behaviour of a Turing machine is determined by its transition function

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\},$$

where

- Q is the set of states,
- Γ is the tape alphabet,
- L and R denote head movements.

The transition function specifies the next action for every state-symbol pair.

Example Computation

Suppose the tape initially contains

1011.

The Turing machine

- reads one symbol,
- updates its state,
- writes if necessary,
- moves the head,
- repeats the process.

Eventually, the machine halts with the desired output on the tape.

Why are Turing Machines Important?

Turing machines provide

- A universal model of computation
- The basis for computability theory
- The foundation of algorithm design
- A reference model for computational complexity

Modern computers are far more practical, but they are computationally equivalent to Turing machines.

From Classical to Quantum Turing Machines

A classical Turing machine processes classical bits.

A quantum Turing machine processes quantum bits (qubits).

Key differences

- Superposition
- Entanglement
- Unitary evolution
- Quantum measurement

These ideas motivate the study of quantum computation.

Summary and Concept Check

Summary

- A Turing machine is an abstract model of computation.
- It consists of a tape, head, finite control, and transition function.
- It executes algorithms through sequential state transitions.
- Turing machines define the notion of computability.
- Quantum Turing machines extend this model using qubits and quantum mechanics.

Concept Check

- 1 What are the main components of a Turing machine?
- 2 What is the role of the transition function?
- 3 Why is the tape considered the memory of the machine?
- 4 How does a quantum Turing machine differ from a classical Turing machine?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the meaning of computational complexity.
- Distinguish between time and space complexity.
- Interpret Big-O notation.
- Describe the complexity classes P, NP and NP-Complete.
- Explain why computational complexity is important for quantum computing.

What is Computational Complexity?

Computational complexity studies the resources required to solve computational problems.

The two most common resources are

- Time (number of computational steps)
- Space (memory required)

The objective is to determine how these resources grow as the input size increases.

Big-O Notation

Big-O notation describes the asymptotic growth of an algorithm.

Common examples

$O(1)$	Constant time
$O(\log n)$	Logarithmic time
$O(n)$	Linear time
$O(n \log n)$	Linearithmic time
$O(n^2)$	Quadratic time
$O(2^n)$	Exponential time

Algorithms with lower growth rates generally scale better.

Complexity Class P

Class P contains problems that can be solved efficiently by a deterministic classical computer.

Examples include

- Sorting
- Matrix multiplication
- Shortest path
- Greatest common divisor

These problems are considered computationally tractable.

Complexity Class NP

A problem belongs to NP if a proposed solution can be verified efficiently by a classical computer.

Examples include

- Sudoku verification
- Boolean satisfiability (SAT)
- Hamiltonian cycle verification

Whether every NP problem can also be solved efficiently remains one of the most important open questions in computer science.

NP-Complete Problems

NP-Complete problems are the hardest problems within NP.

Examples

- SAT
- Traveling Salesperson Problem (decision version)
- Graph coloring
- Clique problem

An efficient algorithm for any NP-Complete problem would efficiently solve all problems in NP.

Why is Complexity Important for Quantum Computing?

Quantum computers do not speed up every computation.

Instead, they offer improvements for selected problems.

Examples

- Integer factorization (Shor's algorithm)
- Unstructured search (Grover's algorithm)
- Quantum simulation

Understanding classical complexity helps identify where quantum algorithms may provide an advantage.

Worked Example

Suppose Algorithm A requires

$$O(n^2)$$

operations,

while Algorithm B requires

$$O(2^n).$$

Question:

Which algorithm scales better for large input sizes?

Answer:

Algorithm A scales much better because polynomial growth is significantly slower than exponential growth.

Summary and Concept Check

Summary

- Computational complexity measures the resources required to solve problems.
- Big-O notation describes algorithm growth rates.
- Class P contains efficiently solvable problems.
- Class NP contains problems whose solutions can be verified efficiently.
- Complexity theory helps explain where quantum computers may offer computational advantages.

Concept Check

- 1 What is computational complexity?
- 2 What does Big-O notation describe?
- 3 How does class P differ from class NP?
- 4 Why is computational complexity important when studying quantum algorithms?

Learning Outcomes

After completing this lecture, you will be able to

- Define the complexity classes P, NP and PSPACE.
- Distinguish between solving and verifying computational problems.
- Explain the significance of polynomial time and polynomial space.
- Describe the relationships among the three complexity classes.
- Recognize why these classes are important in quantum computing.

Complexity Classes

A complexity class groups computational problems according to the resources required to solve them.

The most common resources are

- Time
- Memory (Space)

Today we study three important classes:

P , NP , $PSPACE$.

Class P

Class P consists of decision problems that can be solved in polynomial time by a deterministic classical computer.

Typical examples include

- Graph connectivity
- Minimum spanning tree
- Shortest path
- Greatest common divisor

Problems in P are generally regarded as efficiently solvable.

Class NP

Class NP contains decision problems for which a proposed solution can be verified in polynomial time.

Examples include

- Boolean satisfiability (SAT)
- Hamiltonian cycle
- Traveling Salesperson (decision version)

Whether

$$P = NP$$

remains one of the most important unsolved problems in computer science.

PSPACE contains decision problems that can be solved using polynomial memory, regardless of the running time.

Key idea

Memory is limited, but time may be very large.

Examples include

- Generalized game-solving problems
- Quantified Boolean Formula (QBF)

Relationships Among the Classes

The currently known relationships are

$$P \subseteq NP \subseteq PSPACE.$$

Whether any of these containments are strict remains unknown for P versus NP .

It is known that

$$NP \subseteq PSPACE.$$

Why are These Classes Important?

Complexity classes help us

- Compare computational problems.
- Understand algorithm efficiency.
- Predict computational difficulty.
- Identify where quantum algorithms may offer improvements.

Quantum complexity classes such as BQP are defined by extending these classical ideas.

Worked Example

Consider two computational problems.

Problem A

- Solved efficiently in polynomial time.

Problem B

- Requires exponential time but only polynomial memory.

Question:

Which complexity classes best describe them?

Answer:

Problem A belongs to P .

Problem B may belong to $PSPACE$.

Summary and Concept Check

Summary

- P contains efficiently solvable decision problems.
- NP contains problems with efficiently verifiable solutions.
- $PSPACE$ contains problems solvable using polynomial memory.
- The known relationship is

$$P \subseteq NP \subseteq PSPACE.$$

- Complexity theory provides the foundation for studying quantum computational complexity.

Concept Check

- 1 What distinguishes P from NP ?
- 2 What resource defines $PSPACE$?
- 3 Why is $P = NP$? considered an important open problem?
- 4 How does complexity theory motivate quantum computing?

Tutorial Objectives

After this tutorial, you should be able to

- Distinguish between complexity classes.
- Identify separable and entangled states.
- Recall Bell states.
- Explain practical applications of entanglement.
- Solve introductory conceptual problems.

Quick Revision

Complete the following statements.

- 1 Class P consists of problems that can be solved in _____ time.
- 2 NP problems have solutions that can be _____ efficiently.
- 3 PSPACE is defined using polynomial _____.
- 4 Bell states are examples of _____ states.
- 5 Superposition and _____ are essential quantum resources.

Practice 1: Complexity Classes

Match each problem with the most appropriate description.

- | | |
|------------|------------------------------|
| A. Sorting | 1. Polynomial verification |
| B. SAT | 2. Polynomial-time solution |
| C. QBF | 3. Polynomial-space solution |

Discuss your answers with a partner before revealing the solution.

Practice 2: Identify the State

Determine whether each state is separable or entangled.

$$|00\rangle$$

$$|0\rangle \otimes |1\rangle$$

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Explain your reasoning.

Practice 3: Bell States

Write the four Bell states.

Then answer

- Which Bell states contain correlated outcomes?
- Which contain anti-correlated outcomes in the computational basis?

Discuss how Bell states differ from product states.

Worked Example

A two-qubit system is prepared in

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

If the first qubit is measured as

$$|0\rangle,$$

what is the state of the second qubit?

Answer:

The second qubit is also found in

$$|0\rangle.$$

Practice 4: Think and Discuss

Discuss the following questions.

- 1 Why is entanglement considered a quantum resource?
- 2 Does entanglement allow faster-than-light communication?
- 3 Why are some computational problems difficult for classical computers?
- 4 Can every problem benefit from quantum computation?

Multiple Choice Questions

1. Which class contains efficiently solvable decision problems?

- A. NP
- B. PSPACE
- C. P
- D. EXPTIME

2. Which state is entangled?

- A. $|01\rangle$
- B. $|0\rangle \otimes |1\rangle$
- C. $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$
- D. $|10\rangle$

Summary

Today you practised

- Complexity classes
- Big-O concepts
- Bell states
- Quantum entanglement
- Conceptual reasoning

Suggested Self-Study

- Draw the relationship among P, NP and PSPACE.
- Write all four Bell states from memory.
- Explain in your own words why entanglement is useful for quantum computing.